MIL-I-12659A(Ord)
24 June 1954
SUPERSEDING
MIL-I-12659(Ord)
24 April 1953

# MILITARY SPECIFICATION
# INVERTER, 350 VOLT AMPERE

## 1. Scope

1.1 This specification covers an inverter which is a component of Set, Wiring, M10A1E1 for use with Gun, Twin, 40MM, Self-Propelled, M42.

## 2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND PUBLICATIONS

2.1 The following specifications, and drawings, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

U. S. Army

| | |
|---|---|
| 51 - 7 | Instruments, Fire Control; General Specification Governing the Manufacture and Inspection of |

DRAWINGS

Ordnance Corps

| | |
|---|---|
| D7692245 | Inverter, 350 Volt Ampere |

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

## 3. REQUIREMENTS

3.1 General. - Unless otherwise specified, the requirements common to all fire control instruments as contained in Specification 51-7 shall be mandatory.

3.2 Material. - Material shall be in accordance with the applicable drawings and specifications forming a part of this specification.

3.3 Electrical

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3.3.1 Power rating. - The inverter shall be rated at 350 volt amperes at 30 percent power factor, having 115 volt, 60 cycle output, with 34 d. c. input.

3.3.2 Balance. - The rotor shall be in dynamic balance at all speeds up to and including 25 percent above rated speed. The balancing shall be accomplished by such means as will preclude the possibility of material shifting and destroying rotor balance. Solder shall not be used as a method of balance.

3.3.3 Characteristics. - Each unit shall have the following characteristics when tested at an ambient temperature between 60° F and 90° F at a barometric pressure of 28 to 32 inches of mercury, and a relative humidity not greater than 80 percent.

3.3.3.1 No load output. -

| | |
|---|---|
| Input volts | 24 volts d. c. |
| Input current | 8 amps d. c. (max) |

3.3.3.2 Rated load output (350 volt ampere)

| | | |
|---|---|---|
| Input volts | 24 volts d. c. | ±2 percent |
| Input current | 12 amps d. c. | ± 15 per cent |
| Output volts | 115 volts a. c. | ± 5 percent |
| Output power | 105 watts | ± 10 percent |
| Output frequency | 60 cycles | ± 3 percent |

3.3.3.3 Operation with 18 volt input

| | | |
|---|---|---|
| Input volts | 18 volts d. c. | ± 1 percent |
| Input current | 9.2 amps d. c. | ± 20 percent |
| Output volts | 87 volts a. c. | min. |
| Output power | 22 watts | min. |
| Output frequency | 60 cycles | ± 3 percent |

3.3.3.4 Operation with 30 volts input

| | | |
|---|---|---|
| Input volts | 30 volts d. c. | ± 1 percent |
| Input current | 17 amps d. c. | ± 15 percent |
| Output volts | 143 volts a. c. | ± 5 percent |
| Output power | 170 watts | ± 10 percent |
| Output frequency | 60 cycles | ± 3 percent |

3.3.4 High potential. - There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength of the insulation sufficient to cause failure of the inverter when it is subjected to the test specified in 4.2.1.

3.3.5 Insulation resistance. - All inverters shall have a minimum insulation resistance of 100 megohms when subjected to the test specified in 4.2.2. The insulation resistance of inverters subjected to the temperature rise test shall be measured immediately following that test and shall be not less than 10 megohms.



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3.3.6 Temperature rise.- The temperature rise of the inverter shall not exceed 65° C (117° F) after it has been subjected to 2 hours continuous operation at the rated conditions specified in 3.3.1. Test shall be conducted in accordance with 4.2.3.

3.4 Environmental

3.4.1 Temperature.- The inverter shall meet the requirements of 3.3.3.2 throughout the air temperature range extending from plus 51.5° C (+125° F) (minimum exposure of 4 hours with full impact of solar radiation, 360 BTU/Ft. Sq/Hr.), to minus 31.2° C (-25° F) (minimum exposure of 3 days without benefit of solar radiation). The inverter shall be capable of safe storage within the air temperature range extending from plus 71° C (+160° F) (for periods as long as 4 hours per day) to minus 62.2° C (-80° F) (for periods of at least 3 days duration). Test shall be conducted in accordance with 4.3.2.

3.4.2 Vibration.- The inverter shall not suffer damage or fail to meet requirements of 3.3.3.2 immediately after being subjected to the vibration test specified in 4.3.3.

3.4.3 Shock.- The inverter shall not suffer damage or subsequently fail to meet the requirements of 3.3.3.2 immediately after being subjected to the shock test of 4.4.4.

3.5 Workmanship.- Workmanship shall be in accordance with Specification 51-7.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 General.- Unless otherwise specified, each inverter shall be inspected for conformance with this specification and Specification 51-7. Unless otherwise specified, standard ambient tests shall be performed at a temperature of 60° F to 90° F, at a barometric pressure of 28 to 32 inches of mercury, and a relative humidity not greater than 80 percent.

4.2 Electrical tests

4.2.1 High potential.- All grounded capacitors shall be disconnected during this test. The inverter shall be subjected to a potential of 900 volts rms, with a frequency of 60 cycles per second, for a period of one minute. Three of the first five and one out of every additional 50 or fraction thereof shall be subjected to the test voltage which shall be applied at a rate not exceeding 500 volts (rms) per second. For all other units the test voltage may be applied instantaneously for a period not less than 5 seconds or more than 30 seconds. Insulation resistance test shall be performed immediately after this test.


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4.2.2 Insulation resistance.- All grounded capacitors shall be disconnected during this test. The insulation resistance shall be measured by an approved method using a source of d-c potential of 500 volts.

4.2.3 Temperature rise.- Three of the first five inverters and one out of every additional 50 or fraction thereof shall be tested. . The inverter shall be tested in still air and shall not be mounted in a manner that will cause heat dissipation through the mounting. Immediately after this test the high point potential and the insulation resistance tests shall be repeated. The temperature rise shall be determined as follows:

a. A stop watch shall be started at the instant the unit is stopped.

b. Necessary bridge connections shall be made.

c. A value of resistance which is slightly smaller than the resistance of the winding shall be preset into the bridge. Observe the null indicator of the bridge; and the instant the needle crosses the "zero" position, record the time and the value of the resistance set into the bridge. This procedure shall be followed until five values of resistance versus time after shutdown are recorded. These resistance values shall then be plotted against the corresponding time, and the curve extrapolated to zero time. The value of (R) at zero time shall be used in equation (2) in order to obtain "T". The value "T" shall then be used in equation (1) to obtain "$T_r$".

$$(1)\quad T_r = T - T_a$$

$$(2)\quad T = \frac{R}{r}(234.5 + t) - 234.5$$

Where: $T_r$ = Temperature rise in degrees centigrade

$T$ = Winding temperature in degrees centigrade.

$T_a$ = Ambient temperature in degrees centigrade at end of temperature rise test.

$R$ = Hot resistance of winding in ohms.

$r$ = Cold resistance of winding in ohms.

$t$ = Temperature of winding in degrees centigrade when cold resistance measurements was made.

4.3 Environmental tests

4.3.1 General.- The environmental tests specified shall be limited to those contained in the following sub-paragraph. Three of the first five inverters and one out of every additional 100 or fraction thereof shall be tested.



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4.3.2 Temperature. - The inverter shall be exposed to an air temperature of minus 62.2° C ( -80° F) for a period of 3 days. The air temperature shall then be raised to minus 31.2° C ( -25° F) and after the internal temperature of the inverter has reached equilibrium (approximately 4 hours), it shall be tested for conformance with 3.3.3.2. The air temperature shall be raised to plus 71° C (+160° F) and after the internal temperature of the inverter has reached equilibrium at that temperature, it shall be allowed to remain at that temperature for a period of 4 hours. The air temperature shall be reduced to plus 51.5° C (+125° F) and after the internal temperature of the inverter has reached equilibrium at that temperature the top surface shall be exposed to 360 BTU/Ft. Sq/Hr., for a period of 4 hours. The inverter shall then be tested for conformance with 3.3.3.2.

4.3.3 Vibration. - The inverter shall be mounted using its own mounting provisions, to simulate field use. Each unit tested shall be vibrated successively along each of the three mutually perpendicular directions with accelerations and excursions as shown below. Applied frequency shall be changed at a uniform rate between the limits indicated; the rate of change shall not exceed one cycle per second.

| Frequency Limits | Excursion | Acceleration | Time (Each Direction) |
|---|---|---|---|
| 0 -- 10 cps | 0.060 inch | ---------- | 1 hour |
| 10 -- 20 cps | 0.030 inch | ---------- | 2 hours |
| 20 -- 400 cps | --------- | 4.5 g's | 2 hours |

4.3.4 Shock. - The inverter shall be mounted, using its own mounting provisions, to simulate field use. A shock of 20 g's shall be imparted three times in each direction along each of the three mutually perpendicular directions, making a total of 18 shocks. Each shock impulse shall have a time duration of 11 ± 1 milliseconds; the maximum g value shall occur at approximately 5.5 milliseconds and shall be within ± 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental, from the measured shock pulse.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with instructions forming a part of the contract or furnished by the contracting officer in connection with the contract.

## 6. NOTES

6.1 Ordering data. - Contracts and invition for bids should specify the title, number, date of this specification and type of packing.

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